

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 22

Enable Multi-Region GuardDuty with SecurityAudit Delegated Administration

Amazon GuardDuty | AWS Organizations | SecurityAudit | Threat Detection

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 15, 2026
STATUS	Complete

Executive Summary

Enabled **Amazon GuardDuty** in the approved organization regions **us-west-2** and **us-east-1**, delegated GuardDuty administration to the **SecurityAudit** account, switched into SecurityAudit, enabled GuardDuty for all AWS Organizations accounts, manually added existing accounts where needed, and generated sample findings to review example threat detections.

Business Problem

Centralized CloudTrail provides audit history, but the environment also needs near-real-time threat detection. Amazon GuardDuty analyzes AWS telemetry and produces findings for suspicious behavior such as credential compromise, crypto-mining, command-and-control communication, and unusual activity. Delegating GuardDuty to SecurityAudit reduces the need to use the management account for security monitoring.

Objectives

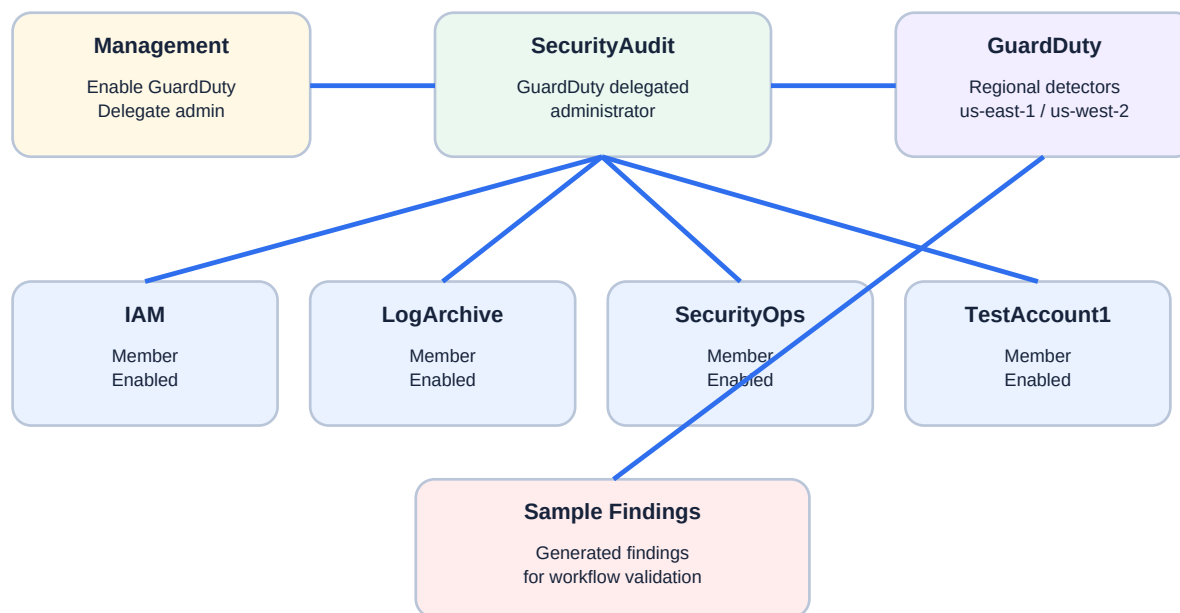
- Enable GuardDuty in us-west-2 and us-east-1.
- Delegate GuardDuty administration to the SecurityAudit account.
- Switch to SecurityAudit and manage GuardDuty from the delegated administrator account.
- Enable foundational GuardDuty for all AWS Organizations accounts.
- Manually add existing accounts so they are enabled, not just future accounts.
- Repeat the setup in both approved regions because GuardDuty is regional.
- Generate sample findings to inspect example threat detections and validate the analyst workflow.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	Amazon GuardDuty
Delegated Administrator	SecurityAudit
Configured Regions	us-west-2 and us-east-1
Foundational GuardDuty	Enabled
Auto-Enable Scope	All AWS Organizations accounts
Existing Member Accounts Added	IAM, LogArchive, SecurityAudit, SecurityOperations, TestAccount1, and one redacted account
Validation	Sample findings generated
Purpose	Centralized multi-account threat detection

Evidence Limitation	Evidence confirms delegated admin, auto-enable, member-account status, and sample findings. Region-specific screenshots do not independently show both regions for every step, so the dual-region completion is documented from the implementation statement and lab workflow.
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Architecture



GuardDuty is enabled per region, then managed centrally from SecurityAudit for member accounts.

Figure 1 - GuardDuty delegated administration and member-account enablement model.

Implementation Summary

1. Signed into the IAM Identity Center access portal and entered the management account with AdministratorAccess.
2. Enabled GuardDuty in us-west-2.
3. Registered SecurityAudit as the GuardDuty delegated administrator.
4. Enabled GuardDuty in us-east-1 and repeated the delegated administration setup.
5. Switched to the SecurityAudit account using the appropriate security administrator access.
6. Opened GuardDuty Accounts in SecurityAudit.
7. Set foundational GuardDuty auto-enable to Enable for all AWS Organizations accounts.
8. Manually selected existing organization accounts and added them as GuardDuty members where required.
9. Repeated account enablement in both us-east-1 and us-west-2.

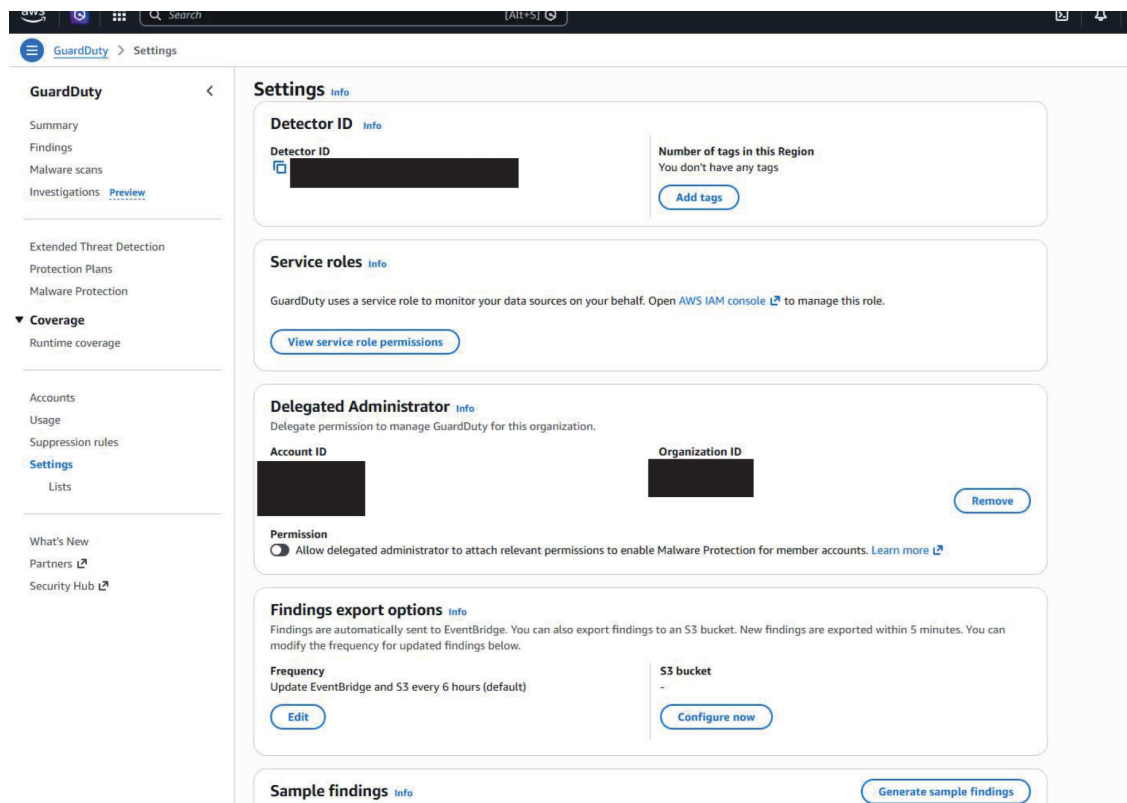
10. Generated GuardDuty sample findings to review example detections and analyst workflow screens.

REGIONAL SCOPE	GuardDuty is a regional service. Enabling it for all accounts in one region does not automatically configure the same setting in every other region.
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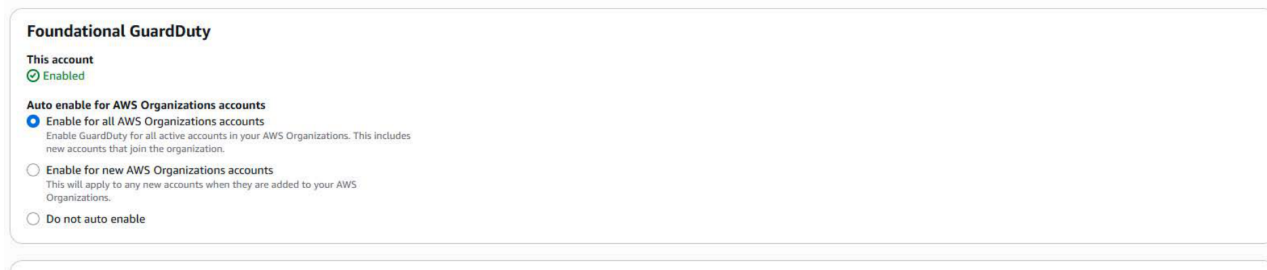
Detection and Coverage Model

Delegated admin	SecurityAudit	Security monitoring administration occurs outside the management account.
Approved regions	us-west-2 and us-east-1	Matches the environment's active-region strategy after RegionLockout.
Auto-enable	All organization accounts	New accounts receive GuardDuty automatically in configured regions.
Existing accounts	Manually added/enabled	Avoids the gap where existing accounts are not enabled by new-account auto-enable alone.
Sample findings	Generated	Lets the user inspect finding details without needing a real incident.

Evidence

Figure 2 - GuardDuty Delegated Administrator Registered

The GuardDuty Settings evidence shows a detector ID, service role section, delegated administrator configuration, and redacted account and organization identifiers. The page also shows the sample findings control.

Figure 3 - Foundational GuardDuty Auto-Enabled for All Organization Accounts

The GuardDuty foundational configuration evidence shows This account: Enabled and the auto-enable option set to Enable for all AWS Organizations accounts.

Figure 4 - Existing Organization Accounts Enabled

Name	Type	Status
TestAccount1	Via Organizations	Enabled
LogArchive	Via Organizations	Enabled
SecurityOperations	Via Organizations	Enabled
	Via Organizations	Enabled
IAM	Via Organizations	Enabled

The GuardDuty accounts evidence shows TestAccount1, LogArchive, SecurityOperations, a redacted account, and IAM with Type Via Organizations and Status Enabled.

Figure 5 - Sample GuardDuty Findings Generated

Findings (404) Info

Create suppression rule Actions

Saved filters

Apply saved filters

Filter findings

Status

Current

Threat type

All findings

< 1 >

Title

[SAMPLE] Potential Kubernetes cluster compromise of eks-demo-cluster indicated by a sequence of actions.

[SAMPLE] Potential credential compromise sequence of actions.

[SAMPLE] Potential data compromise of o of actions associated with IAMUser/john_doe.

[SAMPLE] Potential ECS cluster compromise sequence of actions.

[SAMPLE] Potential EC2 instance group co indicated by a sequence of actions.

[SAMPLE] Potential EC2 instance group co indicated by a sequence of actions.

[SAMPLE] Potential EC2 instance group co indicated by a sequence of actions.

[SAMPLE] Potential EC2 instance group co indicated by a sequence of actions.

[SAMPLE] Potential ECS cluster compromise sequence of actions.

[SAMPLE] Potential EC2 instance group co indicated by a sequence of actions.

Potential Kubernetes cluster compromise of eks-demo-cluster indicated by a sequence of actions.

Critical First seen 2 minutes ago, last seen a minute ago

A sequence of actions involving 4 signals indicating a potential compromise of EKS cluster eks-demo-cluster in account 111122223333 between eventFirstSeen ar eventLastSeen with the following behaviors:

- 3 MITRE tactics observed: Execution, Impact, Privilege Escalation

- 3 MITRE techniques observed: T1204.002 - User Execution: Malicious File, T1496 - Resource Hijacking, T1543 - Create or Modify System Process: Container Serv

- DNS Queries to suspicious domains:

- Cryptomining Domains: crypto.guardduty2activityb.com

- Detected cryptomining processes: xmrig

View details Actions

Is this finding useful? 0 0

Overview Signals (4) Resources (3)

Finding ID

0778da7810d0467aab5482f84f407ff1

Type

AttackSequence:EKS/CompromisedCluster

Region

us-east-1

Account

374198398598

Attack sequence Info

Signals

June 15, 2024, 17:51:16 (UTC-06:00)

A Bitcoin-related domain name was queried by EC2 instance i-00011122233344455.

Additional signals observed (2)

View all

June 15, 2024, 17:51:16 (UTC-06:00)

A privileged container with root level access was launched on an EKS Cluster.

The GuardDuty findings evidence shows 404 sample findings and a selected critical sample finding. The selected example is an EKS compromised-cluster attack sequence in us-east-1 and includes MITRE techniques, crypto-mining indicators, and resource details.

Security Analysis

GuardDuty adds managed threat detection on top of the existing CloudTrail and centralized logging architecture. It produces findings instead of requiring the user to manually inspect every log event.

Delegating administration to SecurityAudit aligns GuardDuty operations with the security monitoring account and reduces routine use of the management account.

Enabling GuardDuty in both us-west-2 and us-east-1 matches the portfolio's approved-region strategy while avoiding coverage gaps in active regions.

Auto-enable for all organization accounts helps future accounts receive GuardDuty automatically. Manually adding existing accounts closes the gap for accounts that already existed before the setting was enabled.

Sample findings are useful for analyst workflow validation, but they are not proof of a real compromise. They should be documented as generated examples.

Lessons Learned

GuardDuty is effectively a managed AWS threat detection service that analyzes telemetry such as CloudTrail, VPC Flow Logs, and DNS-related signals.

GuardDuty must be configured per region. Multi-account does not mean global-by-default.

Delegated administration is a recurring AWS Organizations pattern: the management account performs initial registration, then a purpose-built security account performs routine administration.

Generated sample findings help users learn severity, affected resources, attack sequence, MITRE technique fields, and investigation workflow without creating real malicious activity.

Production Improvements

- Enable GuardDuty in every enabled region for a production organization unless policy intentionally prevents regional usage.
- Send GuardDuty findings to Security Hub, EventBridge, or a SIEM for central triage.
- Create suppression rules only after understanding expected false positives and documenting rationale.
- Evaluate additional GuardDuty protection plans such as S3 Protection, EKS Protection, Malware Protection, RDS Protection, Lambda Network Activity Monitoring, and Runtime Monitoring.
- Create alerts for high and critical findings and define an incident-response workflow.
- Track GuardDuty costs and findings volume after the free trial period.

Skills Demonstrated

Amazon GuardDuty	Delegated administration	AWS Organizations
Multi-region enablement	Member account onboarding	Sample findings

SecurityAudit operations	Threat finding triage	Management-account reduction
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Resume Bullet

RESUME

Enabled Amazon GuardDuty across approved AWS regions, delegated GuardDuty administration to SecurityAudit, enabled protection for all organization accounts, manually onboarded existing member accounts, and generated sample findings to validate the threat-detection workflow.

STAR Interview Story

Situation: The AWS organization had centralized logging, but still needed managed threat detection across accounts and active regions.

Task: Enable GuardDuty using delegated administration and ensure both existing and future organization accounts receive coverage.

Action: I enabled GuardDuty in us-west-2 and us-east-1, delegated GuardDuty administration to SecurityAudit, enabled GuardDuty for all organization accounts, manually added existing accounts, and generated sample findings.

Result: The environment now has multi-region, multi-account GuardDuty coverage managed from SecurityAudit, plus sample findings for validating the threat analysis workflow.

Version History

v1.0.0	July 15, 2026	Initial documented implementation. Enabled GuardDuty in us-west-2 and us-east-1, delegated administration to SecurityAudit, enabled GuardDuty for all organization accounts, manually added existing accounts, generated sample findings, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Enable GuardDuty the Right Way.
- User-provided GuardDuty delegated administrator screenshot.
- User-provided GuardDuty auto-enable screenshot.
- User-provided GuardDuty member account status screenshot.
- User-provided GuardDuty sample finding screenshot.
- Project 20 - Delegated Administration for IAM Identity Center and CloudTrail.
- Project 21 - Security Team Permission Sets in IAM Identity Center.